

SGA 5 – Exposé I

Dualizing Complexes

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Introduction

This exposé is devoted to the biduality theorem in étale cohomology. The theory developed here and in SGA A, Exposé XVIII (where SGA A denotes SGA 4), for the étale topology and constructible torsion sheaves on preschemes, is formally very close to the theory developed in Hartshorne [H] for the Zariski topology and coherent sheaves; the latter served, to a very large extent, as its model. But while, in the case of “continuous” coefficients, the existence of dualizing complexes presents no special difficulty, in the case of “discrete” coefficients it is much more delicate to prove, and seems to require resolution of singularities in an essential way. See, however, Deligne’s biduality theorem (SGA 4^{1/2}, Finiteness Theorem 4.3).

Section 1 contains the definition of dualizing complexes and the formal properties which follow from it, notably the interpretation of the dualizing functors as “exchangers of functors.”

In Section 2 one proves that, on a connected locally noetherian prescheme, a dualizing complex is determined uniquely, up to a translation of degrees and tensoring by an invertible sheaf.

In Section 3 we come to the central theorem of the exposé, Theorem 3.4.1, which asserts that on a “good” regular prescheme X —for example, an excellent noetherian regular prescheme of characteristic zero¹—the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing.

Section 4 gives the applications of the theory to the local duality theorem. This theorem says that, on a “good” prescheme X endowed with a closed point x whose residue field is separably closed, and for a constructible sheaf F , there is a perfect pairing between $\underline{H}_x^i(F')$ and $\underline{H}_x^{-i}(F)$, where F' is the dual of F , that is,

$$F' = R \underline{\mathrm{Hom}}(F, K_X),$$

with K_X a dualizing complex.

Finally, in Section 5, one proves directly that on a regular prescheme of dimension 1 the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing, for n prime to the residual characteristics.

¹This last restriction will become unnecessary once Hironaka-style resolution of singularities and the purity theorem are available for such preschemes.

1. Definition and formal properties of dualizing complexes

Fix a coefficient ring $A = \mathbb{Z}/n\mathbb{Z}$. If X is a prescheme, we denote by A_X the constant sheaf on $X_{\text{ét}}$ with value A , and by $D(X)$ the derived category of the category of A_X -modules. We denote by $D_c(X)$ the full triangulated subcategory of $D(X)$ consisting of those complexes F such that each $H^i(F)$ is a constructible A_X -module. We put

$$D_c^*(X) = D^*(X) \cap D_c(X),$$

where $*$ = $\emptyset, +, -, \text{ or } b$.

Definition 1.1. An A_X -module I is called *quasi-injective* if

$$\underline{\text{Ext}}^i(F, I) = 0$$

for every constructible A_X -module F and every $i > 0$.

Remarks 1.2.

- a) In contrast with what happens for “continuous” coefficients, cf. [H], II.7.17, a quasi-injective sheaf is not in general injective. For example, let $X = \text{Spec}(\mathbb{R})$ and $A = \mathbb{Z}/2\mathbb{Z}$. Then A_X is quasi-injective, but is not injective and is not even of finite injective dimension, since

$$H^*(X; A_X) = H^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}/2\mathbb{Z})$$

is a polynomial algebra on one generator of dimension 1.

- b) Let I be a quasi-injective A_X -module. It is false in general that the relation $\underline{\text{Ext}}^i(F, I) = 0$ for $i > 0$, valid for constructible F , remains valid for all F . Indeed, take $A = \mathbb{F}_\ell$. Let k be a field, let $\bar{k}$ be a separable closure, let

$$f : \bar{x} = \text{Spec}(\bar{k}) \longrightarrow X = \text{Spec}(k)$$

be the canonical morphism, and suppose that, for every connected étale cover $X' \rightarrow X$, there exists a connected étale cover $X'' \rightarrow X'$ such that $[X'' : X']$ is divisible by ℓ ; one may take, for example, $k = \mathbb{Q}$. Consider the exact sequence

$$0 \longrightarrow A_X \xrightarrow{\varepsilon} f_* f^* A_X \longrightarrow F \longrightarrow 0, \tag{*}$$

defined by the adjunction morphism ε . In view of the hypothesis, the section of $\text{Ext}^1(F, A_X)$ defined by $(*)$ does not vanish over any étale U over X ; hence

$$\underline{\text{Ext}}^1(F, A_X)_{\bar{x}} \neq 0.$$

But one checks immediately that A_X is quasi-injective.

Proposition 1.3, cf. [H] 1.7.6. Let X be a prescheme and $N \in \mathbb{Z}$. The following conditions on $K \in \text{Ob } D^+(X)$ are equivalent:

- (i) K is isomorphic, in $D(X)$, to a bounded complex I such that each I^i is quasi-injective and $I^i = 0$ for $i > N$;
- (ii) for every $F \in \text{Ob } D_c^b(X)$ such that $H^i(F) = 0$ for $i < 0$, one has

$$\underline{\text{Ext}}^i(F, K) = 0 \quad (i > N);$$

- (iii) for every constructible A_X -module F , one has

$$\underline{\text{Ext}}^i(F, K) = 0 \quad (i > N).$$

Proof. The implication (i) $\Rightarrow$ (ii) is proved as follows. Let $F \in \text{Ob } D_c^b(X)$ with $H^i(F) = 0$ for $i < 0$, and let $I \in \text{Ob } D^+(X)$ be bounded, with each I^i quasi-injective and $I^i = 0$ for $i > N$. We must show that $\underline{\text{Ext}}^i(F, I) = 0$ for $i > N$. By induction on the number of indices i for which $H^i(F) \neq 0$, using dévissage, one reduces to the case $N = 0$ and I concentrated in degree 0. Then there is a biregular spectral sequence

$$E_2^{pq} = \underline{\text{Ext}}^p(H^{-q}(F), I) \Rightarrow \underline{\text{Ext}}^*(F, I).$$

Since I is quasi-injective, $E_2^{pq} = 0$ for $p > 0$ and all q , hence

$$E_2^{0i} = \underline{\text{Hom}}(H^{-i}(F), I) \xrightarrow{\sim} \underline{\text{Ext}}^i(F, I),$$

and the assertion follows.

The implication (ii) $\Rightarrow$ (iii) is trivial.

For (iii) $\Rightarrow$ (i), we may suppose that K is bounded below and has injective components. Applying the hypothesis to $F = A_X$ gives $H^i(K) = 0$ for $i > N$. We may therefore replace K by an isomorphic complex K' , bounded, with K'^i injective for $i < N$ and $K'^i = 0$ for $i > N$. One then observes that K'^N is quasi-injective, which completes the proof. $\square$

Remark 1.4. The editor does not know whether Proposition 1.3 remains true if, in condition (ii), the hypothesis $F \in \text{Ob } D_c^b(X)$ is replaced by $F \in \text{Ob } D_c^+(X)$.

Definition 1.5. The *quasi-injective dimension* of K , denoted

$$\dim . q. inj(K),$$

is the greatest lower bound, in $\overline{\mathbb{R}}$, of the integers N satisfying the equivalent conditions of Proposition 1.3.

If K has finite quasi-injective dimension, then the functor

$$\mathrm{R} \underline{\mathrm{Hom}}(\cdot, K) : \mathrm{D}(X) \longrightarrow \mathrm{D}(X)$$

induces a functor $\mathrm{D}_c^b(X) \rightarrow \mathrm{D}^b(X)$.

Later we shall see that if X is smooth of dimension d over a field k and n is prime to $\mathrm{char}(k)$, then

$$\dim .q.\mathrm{inj}(A_X) = 2d.$$

The example of Remark 1.2(a) shows that a complex of finite quasi-injective dimension need not have finite injective dimension. Nevertheless one has the following result.

Proposition 1.5.2. Let X be a locally noetherian prescheme and let $K \in \mathrm{Ob} \mathrm{D}^+(X)$. Suppose that there exists an integer N such that $\mathrm{cd}_{\mathbf{n}}(X) \leq N$, where $\mathbf{n}$ is the set of prime divisors of n in the notation of SGA A, Exposé X, 1. Then, for $N' \in \mathbb{Z}$,

$$\dim .q.\mathrm{inj}(K) \leq N' \implies \dim .\mathrm{inj}(K) \leq N + N'.$$

We shall use the following lemma in the proof.

Lemma 1.5.1. Let C be an abelian category satisfying (AB5) and having a family of generators $(U_i)_{i \in I}$. Let $K \in \mathrm{Ob} \mathrm{D}^+(C)$ and $N \in \mathbb{Z}$. The following conditions are equivalent:

- (i) $\dim .\mathrm{inj}(K) \leq N$;
- (ii) for every $i \in I$ and every quotient F of U_i ,

$$\mathrm{Ext}^n(F, K) = 0 \quad (n > N).$$

Proof. It is enough to prove (ii) $\implies$ (i). By a well-known argument, cf. for example [H] 1.7.6, one is reduced to proving that if $K' \in \mathrm{Ob}(C)$ is such that $\mathrm{Ext}^1(F, K') = 0$ whenever F is a quotient of one of the U_i , then K' is injective. Equivalently, if every morphism from a subobject V of some U_i to K' extends to U_i , then K' is injective. This follows at once from the proof of Lemma 1, p. 136, of Tohoku. $\square$

Proof of Proposition 1.5.2. Assume $\dim .q.\mathrm{inj}(K) \leq N'$ and prove $\dim .\mathrm{inj}(K) \leq N + N'$. Since the category of A_X -modules is generated by the sheaves $A_{U,X}$, where U is étale of finite type over X , Lemma 1.5.1 reduces us to proving

$$\mathrm{Ext}^i(X; F, K) = 0 \quad (i > N + N')$$

when F is a quotient of such an $A_{U,X}$, hence constructible by SGA A, Exposé IX, 2.9. Consider the spectral sequence

$$E_2^{pq} = H^p(X; \underline{\text{Ext}}^q(F, K)) \Rightarrow \text{Ext}^*(X; F, K).$$

The hypothesis $\text{cd}_{\mathbf{n}}(X) \leq N$ implies $E_2^{pq} = 0$ for $p > N$, while $\dim . q. \text{inj}(K) \leq N'$ implies $E_2^{pq} = 0$ for $q > N'$. Thus $E_2^{pq} = 0$ for $p + q > N + N'$, giving the desired assertion. $\square$

Remark 1.5.3. The hypothesis of Proposition 1.5.2 is satisfied, with $N = 2d$, if X is a prescheme of finite type and dimension d over a separably closed field k , with n prime to $\text{char}(k)$, by SGA A, Exposé X, 4.

Proposition 1.6. Let $f : X \rightarrow Y$ be a separated quasi-finite morphism of noetherian preschemes, and let $K \in \text{Ob } D^+(Y)$. Then, for $N \in \mathbb{Z}$,

$$\dim . q. \text{inj}(K) \leq N \quad \implies \quad \dim . q. \text{inj}(R^! f(K)) \leq N.$$

Proof. By the Main Theorem of EGA IV, 8.12.6, there is a factorization $f = uf'$, where f' is an open immersion and u is finite. Since

$$R^! f = R^! f' R^! u,$$

we are reduced to treating separately the case of an open immersion and the case of a finite morphism. If f is an open immersion, the assertion is trivial, since every constructible sheaf on X is then induced by a constructible sheaf on Y . Suppose then that f is finite, and let F be a constructible A_X -module. The duality isomorphisms of SGA A, Exposé XVIII, 3.1.9.6 give

$$f_* \underline{\text{Ext}}^i(F, R^! f(K)) \xrightarrow{\sim} \underline{\text{Ext}}^i(f_* F, K).$$

Since $f_* F$ is constructible by SGA A, Exposé IX, 2.14, it follows that if $\dim . q. \text{inj}(K) \leq N$, then

$$f_* \underline{\text{Ext}}^i(F, R^! f(K)) = 0 \quad (i > N),$$

and hence $\underline{\text{Ext}}^i(F, R^! f(K)) = 0$ for $i > N$, by SGA A, Exposé VIII, 5.5. $\square$

Let now X be a prescheme and $K \in \text{Ob } D^+(X)$. We denote by $\underline{D}_K$ the functor

$$R \underline{\text{Hom}}(\cdot, K).$$

We are going to define, for $F \in \text{Ob } D(X)$, a functorial morphism

$$F \longrightarrow \underline{D}_K \underline{D}_K F,$$

cf. [H], V.1.2. The construction works more generally on a ringed topos. We may first

suppose that K is represented by a complex of injectives, so that the symbols R may be suppressed. The identity morphism

$$\underline{\mathrm{Hom}}(F, K) \longrightarrow \underline{\mathrm{Hom}}(F, K)$$

defines, by Cartan's adjunction formula, a morphism

$$F \otimes \underline{\mathrm{Hom}}(F, K) \longrightarrow K,$$

and a second application of the same formula gives

$$F \longrightarrow \underline{\mathrm{Hom}}(\underline{\mathrm{Hom}}(F, K), K).$$

This is the announced morphism, which we shall call the canonical morphism. We say that the pair (F, K) is *bidualizing*, or *satisfies biduality*, or that F is *reflexive relative to K* , if the canonical morphism

$$F \longrightarrow \underline{D}_K \underline{D}_K F$$

is an isomorphism.

From now on, unless expressly stated otherwise, all preschemes under consideration are assumed locally noetherian.

Definition 1.7. Let X be a prescheme. A complex $K \in \mathrm{Ob} \, D^+(X)$ is called *dualizing* if the following conditions hold:

- (i) K has finite quasi-injective dimension;
- (ii) for every $F \in \mathrm{Ob} \, D_c^-(X)$, one has $\underline{D}_K(F) \in \mathrm{Ob} \, D_c^+(X)$;
- (iii) every $F \in \mathrm{Ob} \, D_c^b(X)$ is reflexive relative to K .

Remarks 1.8.

- a) By standard dévissage, truncating F and using the fact that $D_c(X)$ is triangulated, condition (ii) is equivalent to:
 (ii bis) for every constructible A_X -module F and every $i \in \mathbb{Z}$, $\underline{\mathrm{Ext}}^i(F, K)$ is constructible.
- b) Condition (ii bis) implies $K \in \mathrm{Ob} \, D_c^+(X)$. Unlike the example of coherent sheaves, EGA 0_{III}, 12.3.3, the converse is perhaps not automatic. It is true, however, in the “good” cases considered below; for example when resolution of singularities and purity are available on X , in particular if X is an excellent prescheme of characteristic zero.
- c) By dévissage on F , condition (iii) is equivalent to:
 (iii bis) every constructible A_X -module F is reflexive relative to K .

- d) If $K \in \text{Ob } D^+(X)$ satisfies (i) and (ii), respectively if K is dualizing, then $\underline{D}_K$ induces a functor, respectively an equivalence of categories,

$$(D_c^b(X))^\circ \longrightarrow D_c^b(X).$$

- e) If $K \in \text{Ob } D^+(X)$ has finite injective dimension and satisfies (ii), then for every $F \in \text{Ob } D_c(X)$ one has $\underline{D}_K(F) \in D_c(X)$, and $\underline{D}_K$ induces a functor

$$(D_c^+(X))^\circ \longrightarrow D_c^-(X).$$

If, moreover, K is dualizing, every $F \in \text{Ob } D_c(X)$ is reflexive relative to K , and $\underline{D}_K$ induces equivalences of categories

$$(D_c(X))^\circ \xrightarrow{\sim} D_c(X), \quad (D_c^-(X))^\circ \xrightarrow{\sim} D_c^+(X), \quad (D_c^+(X))^\circ \xrightarrow{\sim} D_c^-(X).$$

It is not known whether this conclusion remains valid without the restriction that K have finite injective dimension.

We now turn to the “exchange formulas.” We shall not return to the notion of the tor-dimension of a complex, see SGA A, Exposé XVII, 4.1.9, and also SGA 6, I, 5. Retain only the following point: if X is a prescheme and $G \in \text{Ob } D^-(X)$, the condition $\underline{\text{Tor}}_i(F, G) = 0$ for $i > N$ and all A_X -modules F is equivalent to the same condition with F constructible, by SGA A, Exposé IX, 2.9 and the commutation of tensor product with filtered inductive limits. Thus there is no need to introduce a notion of “quasi-tor-dimension.”

Lemma 1.9, cf. [H] II.4.3.

- a) Let X be a prescheme and let $F, G \in \text{Ob } D_c^-(X)$. One has $F \otimes^L G \in \text{Ob } D_c(X)$ if $F \in \text{Ob } D_c^-(X)$, or if G has finite tor-dimension.
- b) Let $f : X \rightarrow Y$ be a morphism of preschemes and let $G \in \text{Ob } D_c^-(Y)$. Then $f^*(G) \in \text{Ob } D_c(X)$.

Proof. Part (b) follows trivially from SGA A, Exposé IX, 2.4. We prove (a). Either hypothesis on (F, G) implies that the spectral sequence

$$E_2^{pq} = \sum_{q_1+q_2=q} \underline{\text{Tor}}_{-p}(\text{H}^{q_1}(F), \text{H}^{q_2}(G)) \Rightarrow \text{H}^*(F \otimes^L G)$$

is biregular. Hence one is reduced to the case where F and G are concentrated in degree 0, that is, are constructible A_X -modules. The question is local, so one may suppose X noetherian. Then F admits, by SGA A, Exposé IX, 2.7, a left resolution F' in which the F'^i are of the form $A_{U,X}$ with U étale and of finite type over X . We have

$$F \otimes^L G \xleftarrow{\sim} F' \otimes G,$$

and the result follows since the $A_{U,X} \otimes G$ are constructible. $\square$

Notation 1.10. If X is a prescheme, we denote by $D^b(X)_{\text{torf}}$, respectively $D^b(X)_{\text{qinj}}$, the full subcategory of $D(X)$ consisting of objects of finite tor-dimension, respectively of finite quasi-injective dimension.

Proposition 1.11, cf. [H] V.2.6. Let X be a prescheme and $K \in \text{Ob } D^+(X)$.

- a) For $F, G \in \text{Ob } D^-(X)$, respectively for $F \in \text{Ob } D(X)$ and $G \in \text{Ob } D^b(X)_{\text{torf}}$, there is a functorial isomorphism

$$\underline{D}_K(F \otimes^L G) \xrightarrow{\sim} R \underline{\text{Hom}}(F, \underline{D}_K(G)).$$

- b) If K satisfies conditions (i) and (ii) of Definition 1.7, then

$$G \in \text{Ob } D_c^b(X)_{\text{torf}} \implies \underline{D}_K(G) \in \text{Ob } D_c^b(X)_{\text{qinj}}$$

and $\underline{D}_K(G)$ satisfies condition 1.7(ii).

- c) If K is dualizing, then for $F \in \text{Ob } D_c^-(X)$ and $G \in \text{Ob } D_c^b(X)$, respectively for $F \in \text{Ob } D(X)$ and $G \in \text{Ob } D_c^b(X)$ with $\underline{D}_K(G) \in \text{Ob } D_c^b(X)_{\text{torf}}$, there is a functorial isomorphism

$$\underline{D}_K(F \otimes^L \underline{D}_K(G)) \xrightarrow{\sim} R \underline{\text{Hom}}(F, G).$$

Moreover,

$$\underline{D}_K(G) \in \text{Ob } D_c^b(X)_{\text{torf}} \implies G \in \text{Ob } D_c^b(X)_{\text{qinj}},$$

and G satisfies condition 1.7(ii).

- d) If K is dualizing and has finite injective dimension, then the implications in (b) and (c) are equivalences.

Proof. Part (a) is precisely Cartan's adjunction formula, SGA 6, I, 7.4.

For (b), let $G \in \text{Ob } D_c^b(X)_{\text{torf}}$. If F is a variable constructible A_X -module, then the cohomology objects of $F \otimes^L G$ are constructible, by Lemma 1.9, and vanish outside two fixed bounds independent of F . The same therefore holds for the cohomology objects of $\underline{D}_K(F \otimes^L G)$, and (a) gives the assertion.

For (c), the first assertion follows from (a) applied to $\underline{D}_K(G)$ in place of G , together with the reflexivity of G . The second follows by a similar argument to (b).

For (d), prove for instance the converse implication of (b). Suppose that $\underline{D}_K(G) \in \text{Ob } D_c^b(X)_{\text{qinj}}$ and satisfies condition 1.7(ii). Then, as F varies through constructible modules,

the cohomology objects of $\underline{D}_K(F \otimes^{\mathbf{L}} G)$ are constructible and vanish outside two fixed bounds. The same is consequently true for the cohomology objects of $\underline{D}_K \underline{D}_K(F \otimes^{\mathbf{L}} G)$; but

$$F \otimes^{\mathbf{L}} G \xrightarrow{\sim} \underline{D}_K \underline{D}_K(F \otimes^{\mathbf{L}} G)$$

because K is dualizing and of finite injective dimension, cf. 1.8(d). Hence G has finite tor-dimension. The converse implication in (c) is proved analogously. $\square$

Remarks 1.11.1.

- a) The editor does not know whether assertion (d) of Proposition 1.11 is valid under the sole hypothesis that K be dualizing.
- b) One expresses the preceding proposition pictorially by saying that the dualizing functor $\underline{D}_K$ exchanges the functors $\otimes^{\mathbf{L}}$ and $\mathbf{R} \underline{\mathbf{Hom}}$, as well as the notions of finite tor-dimension and finite quasi-injective dimension. The exchange is, however, fully satisfactory only when K has finite injective dimension.

Proposition 1.12. Let Y be a noetherian prescheme, let $f : X \rightarrow Y$ be a morphism of finite type, and let $K \in \text{Ob } \mathbf{D}^+(Y)$. Suppose that n is prime to the residual characteristics of Y , and put

$$K_X = \mathbf{R}^! f(K_Y), \quad \underline{D}_X = \underline{D}_{K_X}, \quad \underline{D}_Y = \underline{D}_{K_Y},$$

in the notation of SGA A, Exposé XVIII, 3.1.

- a) For $F \in \text{Ob } \mathbf{D}^-(X)$ there is a functorial isomorphism

$$\mathbf{R} f_* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathbf{R} f_*(F). \quad (\text{i})$$

If, moreover, K_X and K_Y are dualizing,² then for $F \in \text{Ob } \mathbf{D}_c^b(X)$ there is a functorial isomorphism

$$\mathbf{R} f_! \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathbf{R} f_*(F). \quad (\text{ii})$$

- b) For $F \in \mathbf{D}_c^-(Y)$ there is a functorial isomorphism³

$$\mathbf{R}^! f \underline{D}_Y(F) \xrightarrow{\sim} \underline{D}_X \mathbf{R}^! f(F). \quad (\text{i})$$

If, moreover, K_X and K_Y are dualizing, then for $F \in \mathbf{D}_c^b(Y)$ there is a functorial isomorphism

$$f^* \underline{D}_Y F \xrightarrow{\sim} \underline{D}_X \mathbf{R}^! f F. \quad (\text{ii})$$

²We shall see below that under fairly general conditions K_Y dualizing implies K_X dualizing.

³If the functor $\mathbf{R}^! f$ has finite cohomological dimension, as for example when Y is of finite type over a field (SGA A, Exposé X, 4.3 and Exposé XVIII, 3.1.7), the isomorphism (b)(i) is valid for $F \in \text{Ob } \mathbf{D}_c(Y)$.

Proof. The isomorphism (a)(i) is just the duality isomorphism of SGA A, Exposé XVIII, 3.1.9.6.

The isomorphism (a)(ii) is obtained by applying $\underline{D}_Y$ to both sides of (a)(i), written with argument $\underline{D}_X F$, and using the finiteness theorem SGA A, Exposé XVII, 5.3.6, which asserts that $\mathrm{R}f_*(\underline{D}_X(F)) \in \mathrm{Ob} \, \mathrm{D}_c(Y)$.

The isomorphism (b)(i) is the induction formula of SGA A, Exposé XVIII, 3.1.12.2.

The isomorphism (b)(ii) is obtained by applying $\underline{D}_X$ to both sides of (b)(i), written with argument $\underline{D}_Y F$, and using 1.9(b). $\square$

Scholium 1.12.1. In figurative language, the dualizing functors $\underline{D}_X$ and $\underline{D}_Y$ exchange the usual and unusual direct images, $\mathrm{R}f_*$ and $\mathrm{R}f_!$, and also the usual and unusual inverse images, f^* and $\mathrm{R}^!f$.

Corollary 1.13. Under the hypotheses of Proposition 1.12, suppose in addition that f is proper, and let $F \in \mathrm{Ob} \, \mathrm{D}^-(X)$. If (F, K_X) is bidualizing, then $(\mathrm{R}f_* F, K_Y)$ is bidualizing. Conversely, if f is finite and $(\mathrm{R}f_* F, K_Y)$ is bidualizing, then (F, K_X) is bidualizing.

Proof. Suppose (F, K_X) is bidualizing. Applying $\mathrm{R}f_*$ to the canonical isomorphism $F \xrightarrow{\sim} \underline{D}_X^2 F$, one obtains

$$\begin{aligned} \mathrm{R}f_* F &\xrightarrow{\sim} \mathrm{R}f_* \underline{D}_X \underline{D}_X F \\ &\xrightarrow{\sim} \underline{D}_Y \mathrm{R}f_* \underline{D}_X F && \text{by (a)(i)} \\ &\xrightarrow{\sim} \underline{D}_Y \underline{D}_Y \mathrm{R}f_* F && \text{by (a)(i).} \end{aligned}$$

The courageous reader will convince himself that the isomorphism obtained is the canonical one; this proves the first assertion.

Now suppose f finite and $(f_* F, K_Y)$ bidualizing, since $\mathrm{R}f_* = f_*$ for f finite. We prove that the canonical morphism $F \rightarrow \underline{D}_X^2 F$ is an isomorphism. By Lemma 1.14(a) below, it suffices to show that the induced morphism

$$f_* F \longrightarrow f_* \underline{D}_X^2 F$$

is an isomorphism. But by applying (a)(i) twice, one convinces oneself that this morphism is precisely the canonical isomorphism

$$f_* F \longrightarrow \underline{D}_Y^2 f_* F,$$

which completes the proof. $\square$

Lemma 1.14. Let $f : X \rightarrow Y$ be a finite morphism of preschemes.

- a) For a morphism $u : F \rightarrow G$ in $D(X)$, u is an isomorphism if and only if f_*u is an isomorphism.
- b) For $F \in \text{Ob } D(X)$, one has $F \in \text{Ob } D_c(X)$ if and only if $f_*F \in \text{Ob } D_c(Y)$.

Proof. For (a), since f_* is exact, one may restrict to the case where F and G are concentrated in degree zero. The assertion then follows immediately from the computation of the fibres of f_*F and f_*G , SGA A, Exposé VIII, 5.5.

For (b), again one may restrict to the case where F is concentrated in degree zero. Necessity is known, SGA A, Exposé IX, 2.14. We prove sufficiency: f_*F constructible implies F constructible. The question is local upstairs, hence a fortiori downstairs, so we may suppose Y and hence X affine. By EGA IV, 17.16.6, there exists a finite family (Y_i) of affine subpreschemes of Y , pairwise disjoint and with union Y , such that, if $X_i = X \times_Y Y_i$, the induced morphism $f_i : X_i \rightarrow Y_i$ factors as

$$X_i \xrightarrow{u_i} X'_i \xrightarrow{v_i} Y_i,$$

where u_i is finite radicial and surjective and v_i is finite étale. Put $F|_{X_i} = F'_i$. By the base-change theorem for finite morphisms, SGA A, Exposé VIII, 5.5, one has

$$f_{i*}(F'_i) \xrightarrow{\sim} (f_*F)|_{Y_i},$$

and by SGA A, Exposé IX, 2.8, one is reduced to proving the assertion for (f_i, F'_i) . Put $F'_i = u_{i*}(F_i)$, so that $f_{i*}(F'_i) = v_{i*}(F'_i)$. It is clear from the definition of constructibility, respectively follows from SGA A, Exposé VIII, 1.1, that F_i , respectively F'_i , is constructible if and only if F'_i , respectively $v_{i*}(F'_i)$, is constructible. This proves the claim. $\square$

Corollary 1.15. Let $f : X \rightarrow Y$ be a separated quasi-finite morphism of noetherian preschemes. Let $K_Y \in \text{Ob } D^+(Y)$, and put, as in Proposition 1.12,

$$K_X = R^!f(K_Y), \quad \underline{D}_X = \underline{D}_{K_X}, \quad \underline{D}_Y = \underline{D}_{K_Y}.$$

If K_Y is dualizing, then K_X is dualizing.

Proof. We already know, by Proposition 1.6, that if K_Y has finite quasi-injective dimension, then the same is true of K_X . We show that if K_Y satisfies condition (ii), respectively condition (iii), of Definition 1.7, then K_X satisfies the same condition.

By the Main Theorem, EGA IV, 8.12.6, f factors as $f = f'i$, where i is an open immersion and f' is finite. Since

$$R^!f \simeq R^!i R^!f',$$

one must treat separately the case of an open immersion and the case of a finite morphism. If f is an open immersion, the assertion is trivial, since a constructible sheaf on X is induced by a constructible sheaf on Y , and formation of $R \underline{\text{Hom}}$ commutes with restriction to an open.

Suppose then that f is finite. It follows from Corollary 1.13 that if K_Y satisfies (iii), then K_X satisfies (iii), taking account of the finiteness theorem SGA A, Exposé IX, 2.14. It remains to prove that if K_Y satisfies (ii), then K_X satisfies (ii). Let $F \in \text{Ob } D_c^-(X)$. We must show that $\underline{D}_X(F) \in \text{Ob } D_c^+(X)$. By the finiteness theorem, $f_*F \in \text{Ob } D_c^-(Y)$. Hence, by the hypothesis on K_Y ,

$$\underline{D}_Y f_*F \in \text{Ob } D_c^+(Y).$$

But

$$\underline{D}_Y f_*F \xrightarrow{\sim} f_*\underline{D}_X(F)$$

by Proposition 1.12(a)(i), and Lemma 1.14(b) completes the proof. $\square$

2. Uniqueness of the dualizing complex

In this section we suppose that A is local, that is,

$$A = \mathbb{Z}/\ell^\nu \mathbb{Z},$$

where ℓ is prime.

Theorem 2.1, cf. [H] V.3.1. Let X be a connected locally noetherian prescheme, and let K be a dualizing complex on X in the sense of Definition 1.7. Let also $K' \in \text{Ob } D_c^+(X)$. Then K' is dualizing if and only if there exist an invertible A_X -module L and an integer r , necessarily determined up to unique isomorphism, such that

$$K' \xrightarrow{\sim} K \otimes^L L[r].$$

Proof. Let L be an invertible A_X -module and r an integer, and put $L^\vee = \underline{\text{Hom}}(L, A_X)$. For $F \in \text{Ob } D(X)$ and $G \in \text{Ob } D^+(X)$ there are canonical isomorphisms

$$\text{R } \underline{\text{Hom}}(F \otimes^L L^\vee[-r], G) \xrightarrow{\sim} \text{R } \underline{\text{Hom}}(F, G \otimes^L L[r]) \xrightarrow{\sim} \text{R } \underline{\text{Hom}}(F, G) \otimes^L L[r]. \quad (*)$$

It follows immediately that, if $K' \simeq K \otimes^L L[r]$, then K' is dualizing.

Moreover, if we write $\underline{D}_X = \underline{D}$ and $\underline{D}_{K'} = \underline{D}'$, then for $F \in \text{Ob } D(X)$ the isomorphism $(*)$ gives

$$\underline{D}'F \xrightarrow{\sim} \underline{D}F \otimes^L L^\vee[-r].$$

Since $\underline{D}K \simeq A_X$, one obtains

$$\underline{D}'K \xrightarrow{\sim} L^\vee[-r],$$

which shows that $L^\vee[r]$, hence L and r , are determined essentially uniquely by K' .

Conversely, suppose that K' is dualizing, and put

$$P = \underline{D}'K = \underline{D}'\underline{D}A_X = \mathbf{R}\underline{\mathrm{Hom}}(K, K').$$

We shall prove:

- (a) $K' \xrightarrow{\sim} K \otimes^{\mathbf{L}} P$,
- (b) $P \xrightarrow{\sim} L[r]$ for an invertible A_X -module L .

This will complete the proof of the theorem.

First observe that, by Proposition 1.9(b)(ii), for $F \in \mathrm{Ob}\, \mathbf{D}_c(X)$ there is an isomorphism

$$F \otimes^{\mathbf{L}} P \xrightarrow{\sim} \underline{D}'\underline{D}F. \quad (**)$$

Applying $(**)$ to $F = K$ gives

$$K \otimes^{\mathbf{L}} P \xrightarrow{\sim} \underline{D}'\underline{D}K \xrightarrow{\sim} \underline{D}'A_X = K',$$

which proves (a).

Applying $(**)$ to

$$F = P' = \underline{D}\underline{D}'A_X$$

gives

$$P' \otimes^{\mathbf{L}} P = \underline{D}'\underline{D}\underline{D}\underline{D}'A_X \xrightarrow{\sim} \underline{D}'\underline{D}'A_X \xrightarrow{\sim} A_X.$$

Thus (b) follows from the following lemma. □

Lemma 2.2, cf. [H] V.3.3. Let X be a connected locally noetherian prescheme, and let $P, P' \in \mathrm{Ob}\, \mathbf{D}_c^-(X)$ be such that

$$P \otimes^{\mathbf{L}} P' \xrightarrow{\sim} A_X.$$

Then there exist an invertible A_X -module L and an integer r such that

$$P \xrightarrow{\sim} L[r], \quad P' \xrightarrow{\sim} L^\vee[-r].$$

Proof. The proof has two steps.

First suppose that $X = \mathrm{Spec}(k)$, with k separably closed. One merely repeats the easy proof given in [H]. The essential point is this: by EGA 0_I, 5.4.3, if M and M' are finite-type A -modules such that

$$M \otimes M' \xrightarrow{\sim} A,$$

then M and M' are invertible and $M' \simeq M^\vee$. This is where the hypothesis that A is local is used.

We now pass to the general case. By the special case just treated, for every $x \in X$ there exists an integer $r(x)$ such that

$$P_{\bar{x}} \simeq A_{\bar{x}}[r(x)], \quad P'_{\bar{x}} \simeq A_{\bar{x}}[-r(x)].$$

Equivalently, $H^i(P_{\bar{x}}) = 0$, respectively $H^i(P'_{\bar{x}}) = 0$, except for $i = -r(x)$, respectively $i = r(x)$, and the corresponding nonzero cohomology modules are isomorphic to $A_{\bar{x}}$. It remains to show that the function $x \mapsto r(x)$ is locally constant.

Assume this for the moment. Since X is connected, r is constant, say equal to r_0 . Then

$$H^i(P) = 0 \ (i \neq -r_0), \quad H^i(P') = 0 \ (i \neq r_0),$$

and

$$H^{-r_0}(P) \otimes H^{r_0}(P') \xrightarrow{\sim} A.$$

Put $L = H^{-r_0}(P)$ and $L' = H^{r_0}(P')$. Let $u : \bar{x} \rightarrow \bar{y}$ be a specialization morphism, and let

$$u^* : L_{\bar{y}} \rightarrow L_{\bar{x}}, \quad u'^* : L'_{\bar{y}} \rightarrow L'_{\bar{x}}$$

be the corresponding specialization morphisms. There is a commutative square

$$\begin{array}{ccc} L_{\bar{y}} \otimes L'_{\bar{y}} & \xrightarrow{\sim} & A_{\bar{y}} \\ \downarrow u^* \otimes u'^* & & \downarrow \\ L_{\bar{x}} \otimes L'_{\bar{x}} & \xrightarrow{\sim} & A_{\bar{x}}. \end{array} \quad (*)$$

Choose a basis e , respectively f , of $L_{\bar{x}}$, respectively $L_{\bar{y}}$, and let e' , respectively f' , be the dual basis of $L'_{\bar{x}}$, respectively $L'_{\bar{y}}$, meaning that $e \otimes e'$ and $f \otimes f'$ map to 1. If $u^*(f) = \lambda e$ and $u'^*(f') = \lambda' e'$, the commutativity of $(*)$ gives $\lambda \lambda' = 1$. Hence u^* is an isomorphism, and u'^* is the contragredient isomorphism. By SGA A, Exposé IX, 2.13, it follows that L and L' are locally constant, hence of finite presentation as A_X -modules. Consequently, for every $x \in X$ the canonical map

$$\underline{\mathrm{Hom}}(L, F)_{\bar{x}} \longrightarrow \mathrm{Hom}(L_{\bar{x}}, F_{\bar{x}})$$

is an isomorphism for every A_X -module F . Since the geometric fibres of L are invertible, L itself is invertible. Similarly L' is invertible, and $L' \simeq L^\vee$. Hence

$$P \simeq L[-r_0], \quad P' \simeq L^\vee[r_0].$$

It remains only to prove that r is locally constant. For $m \in \mathbb{Z}$, the condition $r(x) = m$ is equivalent to

$$H^m(P')_{\bar{x}} \neq 0.$$

Thus, by SGA A, Exposé IX, 2.4, the function r is locally constructible. Therefore, to verify that it is locally constant, EGA IV, 1.10.1, allows one to suppose that X is local. Let x be its closed point and U the open complement.

We prove that r is constant. By induction on the dimension of X , we may already suppose that $r(y)$ is constant and equal to r_0 for $y \in U$; the point is to prove $r(x) = r_0$.

Assume the contrary, and derive a contradiction. By hypothesis

$$P|_U \xrightarrow{\sim} L[r_0]$$

with L an invertible A_U -module. After translating degrees we may suppose $r_0 = 0$. On the other hand $r(x) \neq 0$; write $r(x) = a > 0$. Let

$$i : U \rightarrow X, \quad j : x \rightarrow X$$

be the canonical inclusions. Consider the exact sequence

$$0 \longrightarrow i_!(P|_U) \longrightarrow P \longrightarrow j_*j^*P \longrightarrow 0. \quad (**)$$

By hypothesis the only nonzero cohomology objects of P are $H^0(P)$ and $H^{-a}(P)$, and

$$H^0(P) \xrightarrow{\sim} i_!L.$$

It follows that $(**)$ splits naturally, via the map $P \rightarrow H^0(P)$, hence

$$P \xrightarrow{\sim} i_!(P|_U) \oplus j_*j^*P.$$

Therefore

$$\begin{aligned} P \otimes^L P' &\xrightarrow{\sim} i_!(P|_U) \otimes^L P' \oplus j_*j^*P \otimes^L P' \\ &\xrightarrow{\sim} i_!(P|_U \otimes^L P'|_U) \oplus j_*(j^*P \otimes^L j^*P') \\ &\xrightarrow{\sim} i_!A_U \oplus j_*A_x, \end{aligned}$$

which is impossible, since $i_!A_U \oplus j_*A_x$ is plainly not isomorphic to A_X ; already the degree-zero cohomology sheaves are not isomorphic. This proves Lemma 2.2, and hence Theorem 2.1. $\square$

Continuation marker. Continue SGA 5, Exposé I, Section 3, “Existence of dualizing complexes.”